GRADE 3

MATHEMATICS OLYMPIAD

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 90 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully check your name, class, roll number and other required details on the OMR answer sheet.**
- **Do not forget to sign the OMR sheet. Also write your roll number on the question paper booklet.**
- **Use only black or blue ball pen and HB pencil to fill the OMR answer sheet.**
- **Choose the correct answer. Each question has one correct option.**
- **All 50 questions are compulsory and carry one mark each.**
- **Only OMR answer sheets will be evaluated.**
- **There is no negative marking for incorrect answers.**
- **Calculators, smart watches and mobile phones are not allowed.**
- **After the examination, submit both the OMR sheet and question paper to the invigilator.**

SAMPLE PAPER

Q1. A five-digit number is formed using 2, 4, 6, 7 and 9 exactly once. If the largest possible number is subtracted from the smallest possible number formed by reversing the order of the digits, what is the positive difference?

A. 72,729
B. 72,639
C. 72,819
D. 72,909

Q2. In the number 5,08,364, the digit 8 is moved one place to the right without changing any other digit. By how much does the value of the number change?

A. 72,000
B. 80,000
C. 8,000
D. 7,200

Q3. A number pattern begins 7, 15, 31, 63, 127, ____. Instead of simply continuing the pattern, find the next term if each term is obtained by doubling the previous term and then adding the position number of that term.

A. 257
B. 260
C. 261
D. 262

Q4. A three-digit number has 7 in the hundreds place. Its tens digit is 3 less than its hundreds digit, and its ones digit is twice its tens digit. Which number satisfies all the conditions?

A. 742
B. 746
C. 734
D. 764

Q5. Five numbers are 18, 27, 36, 45 and 54. One number does not follow the same divisibility pattern as the others. Which one is it?

A. 27
B. 36
C. 45
D. 54

Q6. A number is increased by 368 and then the result is decreased by 219. The final number is 746. If the original number had first been decreased by 49 instead, what would the final result have been?

A. 548
B. 658
C. 707
D. 756

Q7. If ★ means 'multiply the first number by 4 and then subtract the second number', what is the value of 8 ★ 5?

A. 22
B. 27
C. 32
D. 37

Q8. Which expression has the greatest value?

A. 37 x 6 - 18
B. 29 x 8 - 20
C. 46 x 5 - 17
D. 34 x 7 – 11

Q9. A number leaves remainder 3 when divided by 7. Which of the following could be the remainder when the same number is divided by 5?

A. 0
B. 1
C. 2
D. 4

Q10. A chocolate bar is divided into 12 equal pieces. Neha eats 1/3 of the bar and her brother eats 1/4 of the bar. What fraction of the bar is still left?

A. 5/12
B. 7/12
C. 1/2
D. ¾

Q11. Which fraction lies between 2/5 and 3/5 and has the smallest possible denominator?

A. 3/8
B. 4/9
C. 5/12
D. 7/15

Q12. A box contains 36 chocolates. First, 1/4 of them are given away. Then 2/3 of the remaining chocolates are packed equally into 3 bags. How many chocolates are left outside the bags?

A. 6
B. 9
C. 12
D. 18

Q13. A ribbon is 3 m 25 cm long. Another ribbon is 1 m 85 cm long. After joining them, 40 cm is cut off. What length remains?

A. 4 m 30 cm
B. 4 m 50 cm
C. 4 m 70 cm
D. 5 m 10 cm

SAMPLE PAPER

Q14. Rohan thinks of a number, subtracts it from 1,000, and then divides the result by 7. He gets 86. What number did he think of?

A. 398
B. 402
C. 408
D. 412

Q15. Without multiplying directly, determine the value of 900 x 70 - 900 x 3.

A. 60,300
B. 60,000
C. 63,000
D. 67,300

Q16. A teacher has 144 pencils. She divides them equally among some boxes so that each box gets 12 pencils. She then removes 3 pencils from each box. If 9 pencils remain unused after filling the boxes, how many pencils were originally used in the boxes?

A. 108
B. 117
C. 126
D. 135

Q17. A tank contains 3 litres of water. Each bottle holds 375 mL. After filling as many complete bottles as possible, how much water remains?

A. 0 mL
B. 125 mL
C. 250 mL
D. 375 mL

Q18. A movie starts at 3:45 p.m. It lasts 1 hour 50 minutes, followed by a 20-minute break. At what time does the next programme begin?

A. 5:35 p.m.
B. 5:45 p.m.
C. 5:55 p.m.
D. 6:05 p.m

Q19. A bus leaves at 8:35 a.m. and reaches at 12:20 p.m. It stops for 25 minutes on the way. For how long was the bus actually moving?

A. 3 h 00 min
B. 3 h 10 min
C. 3 h 20 min
D. 3 h 45 min

Q20. A shopkeeper has 8 kg of rice. He sells 2 kg 750 g in the morning and 1 kg 875 g in the evening. How much rice remains?

A. 3 kg 375 g
B. 3 kg 425 g
C. 3 kg 475 g
D. 4 kg 375 g

Q21. A rectangular garden has an area of 96 m² and a length of 12 m. A path of width 1 m is made all around the inside edge. What is the area of the garden left for planting?

A. 60 m²
B. 70 m²
C. 80 m²
D. 84 m²

Q22. Find the missing number: 4, 9, 19, 39, 79, ____.

A. 149
B. 159
C. 160
D. 169

Q23. Find the missing number: 96, 48, 24, 12, 6, ____; then find the sum of the last two terms.

A. 6
B. 7
C. 8
D. 9

Q24. If A = 1, B = 2, C = 3, ..., Z = 26, what is the value of the word MATH if each letter is first doubled and then its original value is added?

A. 96
B. 100
C. 104
D. 108

Q25. If BOOK is coded as 2-15-15-11, and each code number is increased by the position of the letter in the word, how would MATH be coded?

A. 14-2-22-12
B. 14-3-22-12
C. 13-2-21-11
D. 14-3-21-12

Q26. In a queue, Anaya is 11th from the front and 14th from the back. If 4 children leave the queue from positions behind Anaya, how many children are left in the queue?

A. 20
B. 21
C. 24
D. 25

SAMPLE PAPER

Q27. If 1st March is a Tuesday in a non-leap year, what day of the week will 29th March be?
A. Monday
B. Tuesday
C. Wednesday
D. Thursday

Q28. A student has ₹800. She buys a notebook for ₹175, a geometry box for ₹235 and 3 pens at ₹48 each. She also receives a ₹50 discount on the total purchase. How much money does she have left?
A. ₹246
B. ₹296
C. ₹346
D. ₹396

Q29. A child has ₹250 and wants to buy as many identical notebooks as possible at ₹27 each. After buying the notebooks, she must keep at least ₹10. What is the greatest number of notebooks she can buy?
A. 8
B. 9
C. 10
D. 11

Q30. A rectangular playground is 18 m long. Its perimeter is 62 m. A 3 m wide gate is made in one side. What is the length of the fence needed around the playground excluding the gate?
A. 56 m
B. 59 m
C. 62 m
D. 65 m

Q31. A square is divided by two straight lines that cross at its centre, but one line is horizontal and the other is vertical. How many regions are formed? If both lines are extended beyond the square, how many intersection points do the two lines have?

A. 4 regions, 1 point
B. 4 regions, 2 points
C. 8 regions, 1 point
D. 8 regions, 2 points

Q32. A shape has 5 sides. One of its interior angles is a right angle, but the shape is not a rectangle or square. Which description is possible?
A. Triangle
B. Pentagon
C. Hexagon
D. Circle

Q33. Five children P, Q, R, S and T stand in a line. Q is immediately to the right of P. R is between Q and S. T is at the extreme left. Who is in the middle?
A. P
B. Q
C. R
D. S

Q34. A robot moves 7 m north, then 4 m east, then 3 m south, and finally 4 m west. How far and in which direction is it from its starting point?
A. 4 m north
B. 4 m south
C. 7 m north
D. 7 m east

Q35. Three consecutive multiples of 5 have a sum of 120. What is the middle number plus the greatest number?
A. 75
B. 80
C. 85
D. 90

Q36. A number multiplied by 8 gives 216. Instead of multiplying that number by 8, it is multiplied by 5 and then 12 is added. What is the result?
A. 135
B. 147
C. 157
D. 167

Q37. There are 96 students. They are divided equally into groups of 8. Each group is then split into 4 equal teams. How many students are in each team?
A. 2
B. 3
C. 4
D.6

Q38. A three-digit number has these properties: its hundreds digit is 2 more than its tens digit; its ones digit is 3 less than its hundreds digit; and the sum of its digits is 15. What is the number?
A. 654
B. 663
C. 672
D. 681

Q39. Find the next number: 3, 6, 12, 24, 48, 96, ____. Then subtract 18 from it.
A. 174
B. 180
C. 186
D. 192

SAMPLE PAPER

Q40. I am a two-digit number. I am divisible by both 4 and 6. My digits add to 9. Which number am I?

A. 18

B. 24

C. 36

D. 48

Q41. What is the smallest digit that can replace □ so that 5□72 is divisible by 8?

A. 1

B. 2

C. 3

D. 4

Q42. Four children collected 18, 27, 35 and 22 stickers. A fifth child joins them and the average number of stickers becomes 26. How many stickers did the fifth child have?

A. 26

B. 27

C. 28

D. 29

Q43. A class recorded the number of pages read by five students as 18, 25, 14, 31 and 22. One student reads 9 more pages. The class average then becomes 24 pages. How many pages did that student read?

A. 24

B. 28

C. 30

D. 32

Q44. Three boxes contain 24, 36 and 48 marbles. The marbles are redistributed equally among 6 children. How many marbles does each child receive?

A. 16

B. 18

C. 20

D. 22

Q45. A farmer has 168 mangoes. He packs 6 mangoes in each small box and then puts 4 small boxes into each large carton. How many large cartons can he fill completely?

A. 5

B. 6

C. 7

D. 8

Q46. A father is 28 years older than his daughter. Five years from now, the father will be twice as old as the daughter. How old is the daughter now?

A. 18

B. 20

C. 22

D. 23

Q47. The sum of three consecutive odd numbers is 99. What is the product of the smallest and greatest numbers?

A. 1023

B. 1053

C. 1089

D. 1125

Q48. A three-digit number has these properties: its hundreds digit is 2 more than its tens digit; its ones digit is 3 less than its hundreds digit; and the sum of its digits is 15. What is the number?

A. 654

B. 663

C. 672

D. 681

Q49. Find the next number: 3, 6, 12, 24, 48, 96, ____. Then subtract 18 from it.

A. 174

B. 180

C. 186

D. 192

Q50. A bag contains red, blue and green balls. There are twice as many red balls as blue balls and 4 more green balls than blue balls. Altogether there are 28 balls. How many red balls are there?

A. 8

B. 10

C. 12

D. 14

SAMPLE PAPER

SAMPLE PAPER